

AI Reading Pen Product Demo Build Specification

Final technical product design and event handoff

Updated 23 September 2026 | Primary demo language Marathi | Hindi verification pack included

This specification defines a three hour prototype in which a XIAO ESP32S3 Sense captures one printed Devanagari word, a Mac recognizes it, and the system immediately teaches a reviewed reading sequence through audio. The event build should prove the interaction, not universal OCR or automatic literacy instruction.

Final recommendation. Use a low frame rate preview with automatic capture after the target remains sharp and stable for a short dwell. Keep a visible Capture button and Space key as immediate fallbacks. Run OCR locally; accept only high confidence exact matches from a curated demo manifest; play pre-cached, human-reviewed Marathi audio; and keep cloud TTS warm as a recovery path. Hindi is a prepared switchable mode, not an auto-detected language.

Decisions changed after final critique

Earlier assumption	Final decision	Why it changes the build
Manual capture is the baseline	Automatic capture is primary; manual capture remains available	Hands-free capture makes the object feel like a reading pen, while a button or keypress preserves a fast recovery path.
Continuous stream feeds OCR	Preview at low frame rate; OCR one snapshot	The demo needs framing feedback, not continuous recognition. Snapshot OCR is easier to debug and measure.
Marathi and Hindi can be inferred from OCR	The operator selects mr or hi	Both use Devanagari. Script recognition cannot reliably decide the language of isolated words.
Unicode graphemes produce teaching chunks	Unicode protects text integrity; reviewed lexical data defines pedagogy	A grapheme or orthographic syllable is not automatically the unit a teacher would use.
Cloud TTS is generated live for every step	Cache the full demonstration audio; keep Sarvam streaming as fallback	Isolated aksharas can sound unnatural and network startup time can break the first impression.
Target 200 to 700 ms	Measure and target first audio within 1.5 s; stretch goal below 0.8 s	The earlier figure was unbenchmarked and omitted focus, trigger, model warmup, and audio buffering.
PaddleOCR is enough on its own	Benchmark PaddleOCR against Tesseract on the actual cards	Published model metrics do not predict performance on the chosen lens, crop, typeface, and lighting.
Local TTS is a day one dependency	Local TTS is a post-demo experiment	IndicF5 is promising but adds model, reference-audio, and compute setup risk. Ollama is not the deployment path for it.

Product definition

User and promise

The primary user is a child who can recognize some Devanagari letters but needs help blending them into a word. During the event, the child or judge points the camera at a printed card and holds it steady; the system captures automatically. An adult can use the manual control if recovery is needed. The product promise is narrow: point at one word and hear a clear, reviewed way to read it.

Event scope

Must ship	May ship if stable	Explicitly deferred
XIAO camera preview and snapshot	Adjustable dwell and cooldown controls	On-device OCR
Automatic capture with manual fallback	Live Sarvam streaming	Full-page or finger-nearest-word selection
Local one-word OCR	Hindi mode	Child pronunciation scoring
Curated Marathi lesson manifest	Simple meaning or example	Standalone speaker and enclosure
Cached audio and visible lesson state	Stability progress animation	Child voice assessment

Experience and demonstration

What judges should understand in twenty seconds

The camera is a physical pointer. Recognition happens on the nearby Mac for this prototype. The product teaches decoding with approved chunks instead of merely reading a word aloud.

Two minute sequence

1. Show the live camera view and place the Marathi card घर in the target box.
2. Hold the pen steady. The progress indicator completes, capture fires automatically, and the interface shows घर before playing the reviewed teaching prompt.
3. Repeat with झाड to demonstrate a vowel sign, then फुलपाखरू to demonstrate a longer word.
4. If time remains, switch the visible language control to Hindi and scan किताब. Explain that the same script needs a declared language context.
5. Close with the architectural boundary: the immediate reading loop remains deterministic; richer explanations may arrive asynchronously later.

Fallback demonstration

If the camera path fails, upload a saved crop through the same interface. If OCR fails, choose the recognized word from the manifest and continue the teaching sequence. If the network fails, cached audio remains available. Each fallback preserves the product story and should be rehearsed rather than hidden.

System architecture

XIAO preview → Mac quality and stability gate → automatic or manual snapshot → crop and OCR → confidence gate → reviewed lesson → audio

The XIAO firmware serves a preview and a still JPEG. The Mac owns targeting, frame quality checks, stability detection, automatic triggering, preprocessing, OCR, language selection, lesson lookup, audio playback, metrics, and the operator interface. This keeps firmware changes small and allows every stage after capture to be replayed from a saved image.

Component	Responsibility	Must not do during event
XIAO ESP32S3 Sense	Serve camera preview, snapshot, health status	OCR, TTS, child speech analysis
Capture controller	Measure target quality and motion; auto-capture after dwell; lock and rearm; retain manual trigger	Select a lesson before OCR
OCR adapter	Return candidates, confidence, latency, raw image reference	Silently autocorrect a low confidence result
Lesson engine	Normalize exact word and fetch reviewed chunks and prompts	Invent a breakdown for an unsupported conjunct
Audio adapter	Play cached clips, then use live provider if permitted	Block the UI while waiting for enrichment
Optional enrichment	Meaning, example, follow-up activity	Delay the first pronunciation

Hardware and capture

Seed documents a camera web server for the XIAO ESP32S3 Sense and notes that current units may use the OV3660 after the OV2640 was discontinued. Confirm the installed sensor before the event. The board supplies the camera, digital microphone, Wi-Fi, 8 MB PSRAM, and 8 MB flash, but no speaker. Keep audio on the Mac.

Transport. Use the documented HTTP camera server as the starting point. Keep a low frame rate preview for aiming and request one JPEG for recognition. Do not spend the event designing a custom binary or WebSocket image protocol.

Trigger. Make automatic capture the normal path and keep a large Capture button plus Space key as fallbacks. Sample the target region on the Mac, require acceptable text occupancy and sharpness, then require low frame-to-frame motion for a calibrated 600 to 900 millisecond dwell. After capture, lock the trigger until the card leaves or the target changes so the same word does not fire repeatedly.

Auto capture feedback. Show Finding word, Hold steady, Captured, and Move to next word states with a short progress indicator. The user should understand why capture has or has not fired without seeing technical metrics.

Physical setup. Fix the camera height and focal distance with a simple stand, use diffuse front lighting, and mark the card position. Repeatable geometry will improve OCR more than extra model work during a short build.

Resolution. Capture the highest stable JPEG that keeps the word height comfortably above 40 pixels after cropping, then downscale only if the OCR benchmark supports it. Do not lock a resolution before testing the actual camera revision and lens.

Automatic capture controller

1. Sample only the target region from the preview at roughly four to six frames per second. Do not run OCR on every preview frame.
2. Mark the target eligible when it contains a calibrated amount of dark text, passes the blur or sharpness threshold, and is not overexposed.

3. Accumulate stable time while frame-to-frame motion remains below the calibrated threshold. Reset the progress indicator when motion or quality fails.
4. After 600 to 900 milliseconds of continuous eligibility, request one high-resolution snapshot and lock the trigger immediately.
5. Keep the trigger locked while the same card remains present. Rearm only after the target becomes empty or changes substantially for a short period, with a minimum cooldown as a secondary guard.
6. A manual button or Space key bypasses the dwell check but enters the same locked state, preventing duplicate automatic capture immediately afterward.

OCR decision and benchmark

PaddleOCR PP-OCrv5 includes a Devanagari mobile recognizer that lists Hindi and Marathi among its supported languages. Its published Devanagari score is useful for selecting a candidate, but the event decision must come from the supplied card pack photographed through the actual device. Tesseract remains a lightweight independent fallback with separate Hindi and Marathi trained-data files.

Candidate	Role	Decision rule
PaddleOCR Devanagari PP-OCrv5 mobile	Primary candidate	Use if warm recognition succeeds on at least 22 of 24 cards and all four demo cards in the fixed setup.
Tesseract LSTM with mar or hin	Fallback and benchmark	Use if it meets the same demo-card requirement with lower operational risk.
Apple Vision text recognition	Optional local experiment	Do not depend on it until the installed OS reports and proves the needed recognition language on these cards.
Cloud vision OCR	Emergency only	Use only if credentials, privacy statement, and latency are acceptable; never make it the only route.

Confidence and correction policy

- Normalize OCR text to Unicode NFC, trim punctuation and whitespace, and retain the raw result for diagnostics.
- Accept only one exact manifest match above a calibrated confidence threshold. Calibrate the threshold from card results; do not invent a universal value.
- If the top candidate is not an exact manifest item, display Not confident and invite recapture. A manual operator override must be visible in the event log.
- Do not use edit-distance autocorrection without showing it. A plausible wrong word is worse than an explicit retry in a literacy product.

Reading and teaching engine

Unicode extended grapheme clusters prevent combining marks from being torn apart, but Unicode explicitly allows script-specific tailoring and does not define child-facing pedagogy. The engine therefore separates text integrity from instruction. Code may identify orthographic units and marks; a reviewed manifest supplies the chunks and spoken lesson used in the demonstration.

Layer	Output	Authority
Normalization	NFC word and language tag	Unicode rules
Orthographic analysis	Base letters, vowel signs, virama, anusvara, visarga, conjunct spans	Deterministic code with tests
Teaching chunks	Ordered child-facing units	Teacher or fluent-reader review
Speech script	Whole word, chunks, prompt to repeat	Reviewed language content
Meaning and example	Optional asynchronous explanation	LLM or curated content with safety constraints

Demo manifest examples

Language	Word	Reviewed chunks	Lesson class
Marathi	घर	घ + र	simple consonants and schwa behavior
Marathi	झाड	झा + ड	aa vowel sign
Marathi	फुलपाखरू	फु + ल + पा + ख + रू	long word and multiple vowel signs
Marathi	मित्र	मि + त्र	conjunct
Hindi	किताब	कि + ता + ब	pre-base i sign and aa sign
Hindi	बच्चा	ब + च्चा	geminate conjunct
Hindi	स्वतंत्र	स्व + तं + त्र	multiple conjuncts and anusvara

These are event content decisions, not a general phonological standard. A Marathi and Hindi educator should review the final spoken prompts before use with children.

Speech strategy

Use reviewed cached audio for every demonstration word and prompt. For unseen supported manifest items, Sarvam Bulbul v3 is the preferred live provider because its official documentation lists Marathi and Hindi, provides a streaming WebSocket, and supports a pronunciation dictionary. Keep an OpenAI text-to-speech adapter as a provider fallback and test both voices on the exact scripts.

Route	Use during event	Risk control
Cached WAV or PCM	First choice for demo words	Generate and review before doors open; checksum files.
Sarvam Bulbul v3 WebSocket	First live TTS choice	Open connection early, send keep-alive, and pin pronunciations where needed.

Route	Use during event	Risk control
OpenAI speech endpoint	Provider fallback	Request PCM or WAV and start playback from the stream; pre-test Marathi and Hindi pronunciation.
AI4Bharat IndicF5	Post-event offline experiment	Separate setup branch; measure real-time factor, memory, and voice-reference requirements on the target Mac.
macOS system voice	Last-resort diagnostic	Use only after confirming an acceptable installed voice for the selected language.

Avoid synthesizing isolated combining marks or raw letters such as a vowel sign alone. TTS providers are optimized for natural text. Record or synthesize complete phrases such as “झा, झा आणि ड, झाड” and approve the audio as a unit.

Latency and observability

Stage	Event target	Measurement
Stable dwell	600 to 900 ms	continuous eligible target before automatic trigger
Snapshot transfer	p95 below 250 ms	capture request to JPEG received
Crop and preprocess	p95 below 80 ms	JPEG ready to OCR input
Warm OCR	p95 below 500 ms	OCR call start to result
Manifest lookup	below 10 ms	normalized result to lesson
Cached audio start	below 150 ms	lesson selected to audible playback
Trigger to first audio	p95 below 1.5 s	automatic or manual trigger to audible playback
Stable target to first audio	p95 below 2.2 s	stability dwell start to audible playback

Record timings per capture, including cold start separately. Preload the OCR model, audio player, and lesson manifest before the audience arrives. A green Ready state means the camera responds, the OCR model is warm, the demo audio files exist, and at least one TTS provider or the complete cache is available.

User interface

- Large live preview with a fixed central target band and an exposure or blur warning.
- Visible Marathi or Hindi selector that cannot change automatically during a lesson.
- Automatic capture enabled by default, one visible Capture fallback, a Space key shortcut, and clear Finding word Hold steady Captured Speaking and Move to next word states.
- Detected word, confidence, reviewed chunks, and whether audio came from cache or a provider.
- Retry message that states the corrective action: move closer, hold steady, improve light, or show one word.
- Hidden operator drawer for saved-image input, manual manifest selection, and service health. Do not show it during the happy path.

Data contract

The core loop should use one traceable event object. A minimal example follows.

```
{
  "capture_id": "20260923-001",
  "language": "mr",
  "trigger": {"source": "auto", "stable_ms": 742},
  "ocr": {"raw": "झाड", "normalized": "झाड", "confidence": 0.97},
  "lesson_id": "mr_jhaad_v1",
  "chunks": ["झा", "ड"],
  "audio": {"source": "cache", "asset": "mr_jhaad_v1.wav"},
  "timing_ms": {"transfer": 96, "ocr": 214, "first_audio": 486}
}
```

Test plan and release gate

Test	Pass condition	When
Camera health	Twenty consecutive snapshots without restart	Before integrating OCR
Automatic capture	Each demo card fires once after a stable hold; no repeat while it remains in place	Before integrating audio
Trigger rearm	Removing or changing the card rearms capture within one second	Before code freeze
Manual capture	Capture and Space bypass dwell, use the same lock, and never double-fire	Before code freeze
Clean-card OCR	At least 22 of 24 pack items recognized in fixed setup	After each OCR configuration change
Demo words	घर, झाड, फुलपाखरू and किताब each succeed five consecutive times	Before code freeze
Language control	Same-script words never switch language automatically	Once per build
Offline path	All demo words play with Wi-Fi disabled after capture service is already reachable	Final rehearsal
Recovery	Saved image and manual manifest override each work in under ten seconds	Final rehearsal
Cold restart	Operator restores Ready state from written steps in under three minutes	Before event

Three hour build plan

Time	Work	Exit condition
0 to 20 minutes	Inventory camera sensor; start known camera server; fix physical setup	Preview and repeatable card framing

Time	Work	Exit condition
20 to 45 minutes	Implement snapshot ingestion, manual trigger, target quality checks, auto-capture dwell, lock and rearm	Each demo card auto-captures once and manual capture also works
45 to 85 minutes	Warm PaddleOCR; benchmark cards; compare Tesseract only if needed	One OCR route passes all four demo words
85 to 115 minutes	Load manifest, confidence gate, language selector, lesson display	No unsupported word receives an invented lesson
115 to 140 minutes	Integrate cached audio; test Sarvam or OpenAI only as time allows	All demo words speak offline
140 to 160 minutes	Add operator recovery controls and service health	Each fallback works once
160 to 180 minutes	Freeze code and rehearse the exact two minute sequence	Five clean auto-capture runs; manual and saved-image fallbacks also work

Failure ladder

Failure	Immediate action	After the event
Blurry image	Return a specific focus or distance prompt; reposition card	Add calibrated blur threshold and physical guide
Camera Wi-Fi disconnects	Use saved crop; keep the teaching loop live	Add reconnect and local access-point mode
Paddle installation or model fails	Use prevalidated Tesseract environment	Package a locked environment and models
OCR returns plausible wrong word	Reject unless exact manifest match and threshold passes	Add candidate comparison and domain lexicon
TTS provider stalls	Play cached audio	Add circuit breaker and provider metrics
Conjunct breakdown is disputed	Use reviewed demo manifest wording	Run educator review and version content
Auto-capture does not fire	Use Capture or Space immediately; continue without changing the explanation	Tune target quality and motion thresholds from recorded traces
Auto-capture repeats	Disable auto mode and use manual capture	Strengthen lock, scene-change rearm, and cooldown tests
Hindi or Marathi ambiguity	Use the visible language selector	Use session context, not script-only inference

Privacy safety and accessibility

- Do not record a child's face or voice for the event. Aim at printed cards and keep the microphone out of scope.
- Store test frames locally and delete them after evaluation unless consent and retention rules are defined.
- Disclose that the voice is synthetic when cloud TTS is used.
- Use large high-contrast text and do not rely on color alone for state or language.
- Do not present automated decomposition as a substitute for teacher instruction. Version and review lesson content.

Post-event roadmap

1. Collect a consented, camera-specific corpus across books, fonts, sizes, lighting, and page curvature; report word accuracy by condition.
2. Expand the reviewed lesson manifest with educator input and separate Marathi and Hindi pedagogy where pronunciation differs.
3. Add finger-nearest-word targeting only after word detection is reliable on full-page layouts.
4. Benchmark offline IndicF5 and other local TTS options on the exact Mac; compare startup, real-time factor, memory, and intelligibility.
5. Prototype an onboard speaker and explicit trigger after the laptop loop is stable; do not attempt full Devanagari OCR on the ESP32 without a measured feasibility study.
6. Add child speech assessment only with consent, age-appropriate feedback, and a validation plan for accent and dialect variation.

Sources

[Seeed Studio XIAO ESP32S3 camera usage](#)

[Seeed Studio XIAO ESP32S3 getting started](#)

[PaddleOCR PP-OCRv5 multilingual recognition](#)

[PaddleOCR text recognition module](#)

[Tesseract best trained data](#)

[Sarvam Bulbul model](#)

[Sarvam streaming text to speech](#)

[Sarvam pronunciation dictionary](#)

[OpenAI audio API](#)

[AI4Bharat IndicF5](#)

[Unicode text segmentation](#)